

2(a)	(total volume of molecules is) negligible	M1
	compared with volume occupied by the gas	A1
2(b)(i)	$pV = NkT$	C1
	$4.60 \times 10^5 \times 2.40 \times 10^{-2} = N \times 1.38 \times 10^{-23} \times (273 + 23)$	C1
	or	
	$pV = nRT$	(C1)
	$4.60 \times 10^5 \times 2.40 \times 10^{-2} = n \times 8.31 \times (273 + 23)$ $n = 4.49 \text{ (mol)}$ $N = nN_A$ $= 4.49 \times 6.02 \times 10^{23}$	(C1)
	$N = 2.7 \times 10^{24}$	A1

PUBLISHED

Question	Answer	Marks
2(b)(ii)	volume of one atom = d^3 ($= 2.7 \times 10^{-29} \text{ m}^3$)	C1
	volume of all atoms = $2.7 \times 10^{-29} \times 2.7 \times 10^{24}$	C1
	$= 7 \times 10^{-5} \text{ m}^3$	A1
	or	
	volume of one atom = $(4/3)\pi r^3$ ($= 1.41 \times 10^{-29} \text{ m}^3$)	(C1)
	volume of all atoms = $2.7 \times 10^{24} \times 1.41 \times 10^{-29}$	(C1)
	$= 4 \times 10^{-5} \text{ m}^3$	(A1)
2(c)	numerical comparison between answer to (b)(ii) and $2.4 \times 10^{-2} \text{ (m}^3\text{)}$ showing (b)(ii) is <u>much</u> less than $2.4 \times 10^{-2} \text{ (m}^3\text{)}$	B1